

A novel informer-time-series generative adversarial networks for day-ahead scenario generation of wind power

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HIGHLIGHTS

- A novel generation model for wind power based on *Probspare* self-attention and TimeGANs is developed.
- A method for generating day-ahead wind power scenarios considering numerical characteristics of prediction errors is proposed.
- A joint loss function is designed by combining auxiliary classification supervised training and unsupervised adversarial training.

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ABSTRACT

With the rapid growth of wind power penetration, its inherent stochasticity and uncertainty will seriously affect the stable operation of power systems. How to effectively characterize the uncertainty of wind power is a great challenge for day-ahead power system dispatching, scenario generation is an important method to describe the uncertainty of wind power. Currently, most of the wind power scenarios are generated using a generative adversarial network with two-dimensional convolution as the main structure, which may make it difficult to adequately characterize the temporal features, the day-ahead mode properties, and seasonality of wind power. In this paper, we first establish an auxiliary classification time-series generation adversarial network based on error stratification, construct the numerical characteristic conditional labels that can reflect the fluctuation characteristics of day-ahead wind power and power output level, and design the temporal embedding function that captures the seasonal characteristics of wind power. On this basis, to fully extract the dynamic variation characteristics and global effective information of wind power prediction error sequences, Informer is combined with a time-series generative adversarial network, and a joint loss function incorporating supervised learning and unsupervised learning is constructed. Subsequently, the generated set of prediction error sequences is superimposed with the day-ahead predicted value of wind power to obtain the day-ahead wind power scenario set. Finally, to verify the effectiveness of the proposed method, two datasets from different geographic locations are used to comprehensively evaluate the generated day-ahead wind power scenario set in terms of three aspects: temporal correlation characteristics, fluctuation characteristics, and accuracy. The experimental results indicate that the scenario generation method proposed can improve the quality of the day-ahead wind power scenario set and has an excellent performance in describing wind power uncertainty compared with other methods.

1. Introduction

As the penetration of wind power and other renewables in the power system continues to increase, its random and intermittent nature makes the uncertainty of the power grid increase, bringing significant challenges to power system operation and planning. The scenario analysis method is an effective tool to describe renewables uncertainty, and its essence is to probabilistically model a random vector with a continuous

probability distribution and discretize the probabilistic model into a series of time-series scenarios [1], transforming the stochastic optimization problem into a deterministic problem to deal with [2], offering power system dispatch and operation control decision makers with clear and continuous scenarios that may occur in the future [3], which can provide more quantitative information about renewables uncertainty compared with deterministic methods [4].

The main application situations of the scenario analysis method in

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power systems are power system planning, power market trading, and economic dispatch. Where, the day-ahead scenario analysis is primarily used in the finer time-scale day-ahead economic dispatch [5], unit combination optimization, and day-ahead market trading, which can construct more accurate scenarios to make the power system stochastic optimization problem closer to the optimal solution compared with the medium- and long-term scenario analysis and short-term scenario analysis. Most of the current research on scenario generation has focused on stochastic scenario generation and less on day-ahead time-scale wind power scenario generation. Stochastic scenario generation does not specify the time scale, the generated scenarios only need to meet the overall distribution characteristics of historical wind power data, and do not need to satisfy the specific output characteristics under a certain date. However, the power change rule of day-ahead wind power scenarios needs to match the pattern characteristics belonging to the time period as much as possible, such as the power level, fluctuation, ramping and seasonality, and other landmark characteristics of wind power. Therefore, how to generate accurate and reliable wind power scenarios under the corresponding mode characteristics is still a challenging problem [6].

The majority of previous wind power scenario generation studies have used statistical methods, which require first building a specific mathematical model to describe the probability distribution of wind power, and then constructing the scenario set by random sampling methods such as Monte Carlo Sampling (MCS) or Latin Hypercube Sampling (LHS). The primary probability distribution modeling methods include the Markov Chain (MC) [7], Gaussian process [8,9], time series model [10], and copula function [11–13]. The generative models based on statistical methods are highly interpretable, yet the statistical assumptions that fit the probability distributions are specific expressions. In practical applications, it is difficult to determine an adequate mathematical model that can be applied to many types of wind farms. Moreover, the mathematical model often has the problem of complex parameter solutions. Therefore, the applicability and scalability of the above methods are limited.

With the development of artificial intelligence technology, deep generative models have been widely used in scenario generation. Deep learning-based scenario generation methods mainly comprise Variational Auto-Encoder (VAE) [14], Normalizing Flows (NF) [15], and Generative Adversarial Nets (GAN) [16] models. Such methods do not need to determine the probability distribution model of the data in advance, but extract features directly from the input data, which has muscular flexibility and scalability [17,18] improved the objective function of VAE to address its lack of generalization capability. Nevertheless, VAE is unable to reason about the log-likelihood function of the actual samples accurately and can only approximate the lower bound of the maximum log-likelihood, leading to shortcomings in the quantitative fitting of probabilistic features for the time-series scenario set it generates [19]. introduced NF into scenario-based probabilistic prediction and demonstrated the effectiveness of generating scenarios regarding both predictive worth and predictive quality. The NF model solves the maximum likelihood function of a sample directly through a reversible network with bijective properties. However, this property also causes it to be unable to achieve dimensionality reduction mapping of time-series data, which results in excessive demand for computational resources by the model. Therefore, NF has been applied the least among the three types of models.

GAN avoids the difficulty of directly solving the maximum likelihood function by alternating the training of two deep neural networks, the generator, and the discriminator [20]. applied GAN to renewable energy scenario generation for the first time, and the results showed that GAN can be quickly extended to generate diverse renewable energy scenarios [21]. combines federation learning with Least Squares Generative Adversarial Networks for renewable energy random scenario generation in a privacy-preserving manner [22]. improved the performance of GAN for scenario generation by mitigating the problem of model overfitting

by adding Lipschitz constraints to the discriminator network [23,24]. have applied the GAN-based scenario generation method to the study of stochastic optimal scheduling and demonstrated that GAN is able to accurately fit the probability distribution of renewable energy samples for generating high-quality scenario sets.

Nevertheless, the above GAN model was originally proposed for image generation. Its network structure does not fully consider the characteristics of time series, which makes it hard to capture the time-series dependence of the joint probability distribution for time series data. In response to this problem, there have been studies proposing some GAN models specifically for processing time series [25]. Such as [26], which applies the generative adversarial network architecture directly to renewable time series data by instantiating the generator and discriminator as a recurrent neural network (RNN) to accommodate the generation of time series [27]. adapted the sequence generative adversarial network (SeqGAN) to scenario generation for day-ahead scheduling of microgrids, and the model combines long short-term memory networks and reinforcement learning for capturing temporal correlation. However, the above methods only rely on the binary adversarial game between the generator and the discriminator for learning, and directly model the overall distribution of the data by adversarial training objectives, without considering the conditional distribution of the leaps between the time steps of the renewables time series, which cannot ensure that the network accurately captures the time-series dynamic properties of the data. [28]proposed the time-series generative adversarial networks (TimeGAN) framework, where the model combines the supervised training of an autoregressive model and the flexible framework of GANs to extract the stepwise dynamic properties between each time step of the sequence, showing excellent performance in the task of generating time series.

Different blocks in the TimeGAN framework have different functions and need to solve different problems. However, all blocks of the original TimeGAN are implemented using RNN and the serial structure of the RNN sequence model needs to deal with each element of the sequence sequentially, which is prone to problems such as gradient vanishing when solving long-time sequence tasks. This will lead to further improvement in the quality and efficiency of TimeGAN in realizing the task of wind power scenario generation. In addition, although TimeGAN establishes the mapping relationship between historical data and stochastic vectors, due to the uncontrollability of the model, it is difficult to generate effective day-ahead wind power scenarios that conform to the fluctuating characteristics and seasonality of wind power on a given date.

Therefore, this paper proposes an Informer-TimeGAN day-ahead wind power scenario generation method that considers both forecast information and temporal information. An error stratification block based on the day-ahead power prediction information and a temporal embedding function mining seasonal characteristics are established to ensure the controllability of the generated scenarios. A temporal convolutional network that ensures the temporal coherence of the generated sequences and has parallel computation capability. As well as a self-attention mechanism model, Informer, which accurately captures the key information that determines the classification of the sequences, are embedded within the model to facilitate the fast and effective learning of the historical power data distributions by the model. The main contributions of this paper are summarized as follows:

- (1) A method for day-ahead wind power scenario generation considering ramping characteristics and mean levels of predicted power is proposed. The model training label vectors are constructed by combining prediction error stratification block and Auxiliary Classifier GAN (ACGAN). Further embedding the seasonal information of the sequences to supervise the model in accurately generating error sequences under the corresponding pattern characteristics.

- (2) A novel scenario generation model, Informer-TimeGANs, is established. The stepwise dynamic conditional probability distribution of wind power prediction error sequences is captured by an autoregressive temporal convolutional network. In addition, Informer mines key features of real and generated long-time wind power prediction error sequences based on the *ProbSparse* self-attention layer, motivating the generator as it learns deeper sequence properties during the adversarial game training process.
- (3) A joint loss function is designed by combining autoregressive supervised training, auxiliary classification supervised training, and unsupervised adversarial training, which enables Informer-TimeGANs to deeply excavate the hidden information of real sequences and label vectors. Finally, the strong robustness of the Informer-TimeGANs model is verified by testing on a real dataset.

The remaining sections of this paper are organized as follows, [Section 2](#) introduces the architecture and model details of Informer-TimeGANs, and [Section 3](#) presents a case study of the model proposed in the paper with two real datasets.

2. Informer-TimeGANs for day-ahead wind power scenario generation

2.1. The general framework for wind power prediction error scenario generation

Compared with wind power stochastic scenario generation and medium- and long-term scenario generation, day-ahead scenario generation imposes more stringent requirements on the accuracy of the scenarios themselves. The day-ahead scenario set should be close to the prediction value while accurately reflecting the fluctuation of wind

power uncertainty [29], so this paper generates the set of day-ahead wind power prediction error sequences through Informer-TimeGANs for multiple wind farms simultaneously, and then superimposes the predicted values to generate the day-ahead wind power scenario set. [Fig. 1](#) illustrates the general framework for the day-ahead wind power prediction error scenario generation based on Informer-TimeGANs, which consists of three blocks: Error stratification, Reconstruction, and Adversarial block.

Firstly, the Error stratification block stratifies the wind power prediction error series by the duration of the daily prediction power ramp event, and the power mean level, obtains the numerical distribution characteristics of the errors in different strata, and gets the error series with stratification labels. Next, the Value embedding function and Temporal embedding function in the Reconstruction block map the error sequences set to the latent space, while amplifying the critical seasonal features of the error sequences to obtain the low-dimensional error feature set H_n . H_n can be restored to the original feature space by the Recovery function. Then, in the Adversarial block, the generator generates the error scenario feature set $\hat{H}^n$ under the corresponding label in a targeted manner. The discriminator adopts Informer to distinguish the real or generated error feature set and judge whether the data features satisfy the distribution mapping relationship under the corresponding label. Finally, the Recovery function in the Reconstruction block takes as input the error scenario feature set generated by the generator and outputs the error sequence scenario set $\hat{Err}_m^n$ under the original feature space.

2.2. Error stratification block

Suppose there are N wind farms in a region, and the power prediction error sample data of the n -th wind farm $Err^n = \{e_1^n, e_2^n, \dots, e_T^n\}$, T is the

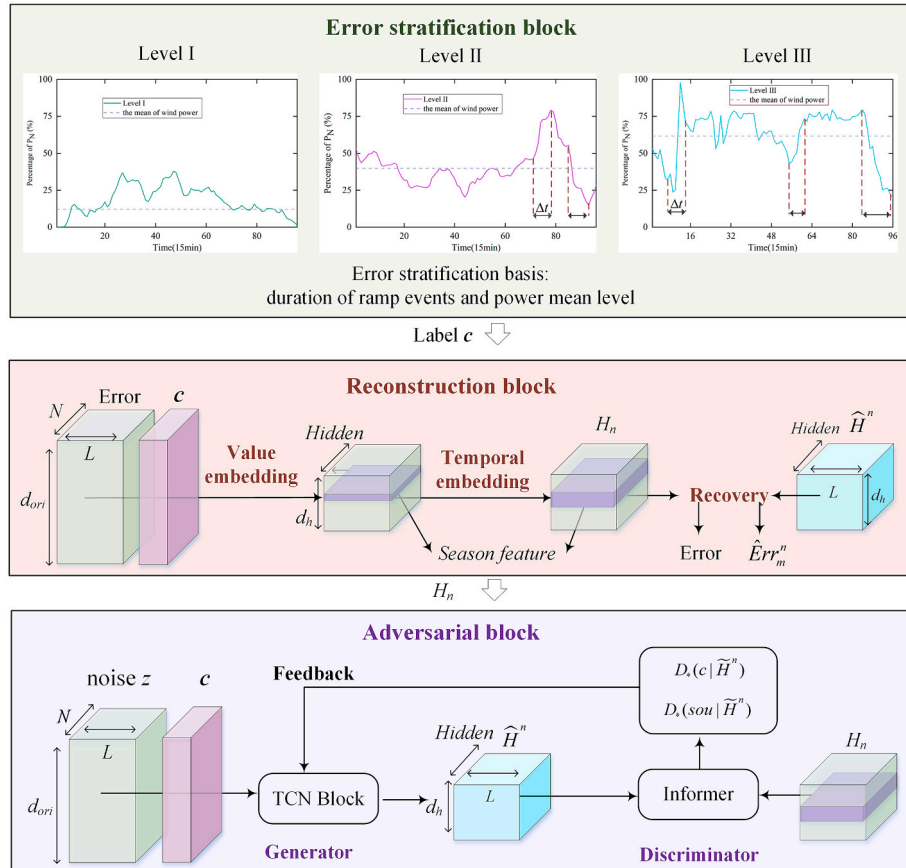


Fig. 1. Overall framework of Informer-TimeGANs.

total length of the error sample. Referring to [28], in this paper, the feature set of Err^n is considered to consist of static features (i.e., features that do not change over time, such as the geographic location of the wind farm and the resource characteristics of the region) and dynamic features (i.e., features that change over time such as seasons and weather). Defining s and x denotes the high-dimensional static and dynamic feature vectors of Err^n data, respectively. Using $\{(s^n, x^n_{1:L})\}_{N=1}^N$ to characterize the Err^n data at moment t , then the day-ahead prediction error sequence $Err^n d$ can be expressed as $\{(s^n, x^n_{1:L})\}_{N=1}^N$, where L represents the length of the day-ahead prediction error sequence.

The prediction error characteristics are closely related to the power fluctuation level and amplitude. The wind power fluctuation is drastic when the ramp event occurs, leading to a large wind power prediction error, while the magnitude of the prediction error value is also proportional to the power mean value level, so the prediction error under different scenarios presents various numerical characteristics. In this paper, to realize the targeted modeling of day-ahead wind power prediction error series under different scenarios, the day-ahead prediction error $Err^n d$ is stratified according to the ramp characteristics and power level. The definition of the ramp event is given in [30]: A ramp event occurs when the absolute value of the difference in power between the first and last moments of the time period $(t, t + \Delta t)$ exceeds a given threshold.

$$|P_{t+\Delta t} - P_t| > P_{\text{threshold}} \quad (1)$$

where P_t and $P_{t+\Delta t}$ are the wind power values at moments t and $t + \Delta t$, $P_{\text{threshold}}$ is the set power threshold. Referring to [31,32] sets $P_{\text{threshold}}$ and Δt to $25\% \cdot P_N$ and 1 h, respectively, P_N is the installed capacity of the wind farm.

Multiple up-ramp and down-ramp events may occur in the day-ahead time scale, and it is not possible to simply stratify $Err^n d$ by the types of ramp events that happen in a day. Thus the total duration of the day-ahead ramp events and the power level are integrated as the basis for the stratification of $Err^n d$. A typical wind farm WF1 is used as the sample data for analysis. The values of the total duration of ramp events in a day are distributed in the interval [0 h, 6.25 h], which is divided into three intervals T1 [0, 1.5 h], T2 (1.5, 3 h], T3 (3, 6.25 h], and the mean value of wind power is divided into P1 [0, 15% P_N], P2 (15% P_N , 40% P_N], and P3 (40% P_N , P_N). The samples are uniformly divided into Level I, Level II, and Level III according to the values of the total duration of the day-ahead ramp event and the power mean level, with $c \in C$ noted as the label vector of the day-ahead prediction error $Err^n d$ used to characterize the data stratification results, and the detailed division is shown in Table 1.

Fig. 2 gives the probability distribution of the prediction error from wind farm WF1. It can be found that the probability distribution differs significantly between the prediction error samples of different levels, and the overall probability density distribution of the data disperses toward the large error interval as the error level rises, which proves the reasonableness of the error stratification in this paper.

2.3. Reconstruction block

The static features s^n and dynamic features $x^n_{1:L}$ of the day-ahead prediction error sequence $Err^n d$ are high-dimensional vectors whose dimension is d_{ori} , where d_{ori} is a variable that keeps changing with time, i.e., the dimensionality of the features varies for the data at different

moments in the sequence. The existence of redundant features in s^n and $x^n_{1:L}$, and the high complexity of the high-dimensional vectors will make the generative model more difficult to learn. Therefore, in this paper, we use the Reconstruction block to map s^n and $x^n_{1:L}$ into the low-dimensional latent space H , obtaining the low-dimensional feature set $H^n = \{h^n_s, h^n_{1:L}\}_{N=1}^N$ of error with dimension d_h , where, $d_h < d_{ori}$. The Reconstruction block contains three components, Value embedding, Temporal embedding, and Recovery function, whose detailed structure is shown in Fig. 3. This section introduces the process of creating the Reconstruction block.

(1) Initial low-dimensional feature extraction of $Err^n d$.

According to the prediction error sequence $Err^n d$ with label c output by the Error stratification block, the Reconstruction block eliminates the redundant features in the $Err^n d$ data by the Value embedding function and establishes the mapping relationship between the original high-dimensional static, dynamic features s^n and $x^n_{1:L}$ of $Err^n d$ and the low-dimensional static, dynamic features h^n_s and $h^n_{1:L}$ to extract the initial low-dimensional feature set, where the dynamic feature x^n_t mapping at time t takes into account the information of the low-dimensional dynamic features h^n_{t-1} at the previous time.

(2) Integration of seasonal features.

Usually, the wind power output and fluctuation levels in spring and winter are larger than those in summer and autumn. Temporal embedding in Fig. 3 shows the mean change curve for the absolute value of daily prediction error in spring, summer, autumn, and winter. It can be seen that the wind power prediction errors are affected by the seasonal change pattern of wind power showing prominent seasonal characteristics. So the Reconstruction block encodes the timestamp $stamp$ of $Err^n d$, taking advantage of the Temporal embedding function to extract the seasonal features of the sequence, and fuses the initial low-dimensional feature set output by Value embedding with the seasonal features to work the final low-dimensional feature set H^n , which will be expressed as

$$\begin{aligned} h^n_s &= V_s(s^n) \\ h^n_t &= V_x(h^n_s, h^n_{t-1}, x^n_t, c) + Temp(stamp) \\ H^n &= \{h^n_s, h^n_t, t = 1, 2, \dots, L\}_{n=1}^N \end{aligned} \quad (2)$$

where $stamp$ is the timestamp of the error sequence, V_s and $Temp$ are the Value embedding function and Temporal embedding function, respectively.

(3) Low-dimensional feature reconfiguration of error.

If the low-dimensional feature set generated in the latent space H directly constitutes the error scenario set will result in a relatively homogeneous generated scenario set. Therefore, the Reconstruction block takes the Recovery function to reconstruct the low-dimensional feature set that is learned by the Adversarial block into the original feature space. During model training, the Recovery function learns the mapping relationship between the low-dimensional features and the original features using the real low-dimensional features H^n of $Err^n d$ as input, so that it can recover accurately from the low-dimensional potential representation to the original error data. At model testing, the generated low-dimensional feature set is input to the Recovery function to reconstruct it into the final error scenario set $\hat{Err}_m^n$ generated by Informer-TimeGANs, where m is the number of generated scenarios. The Recovery function can be expressed as follows:

$$\begin{aligned} \tilde{s}^n &= R_s(\tilde{h}^n_s) \\ \tilde{x}^n_t &= R_t(\tilde{h}^n_t) \end{aligned} \quad (3)$$

Table 1
Details of error stratification.

c	The total duration of ramp events and the power mean interval
Level I	(T1, P1)
Level II	[(T1, P2) & (T2, P1) & (T2, P2) & (T2, P3)]
Level III	[(T3, P2) & (T3, P3)]

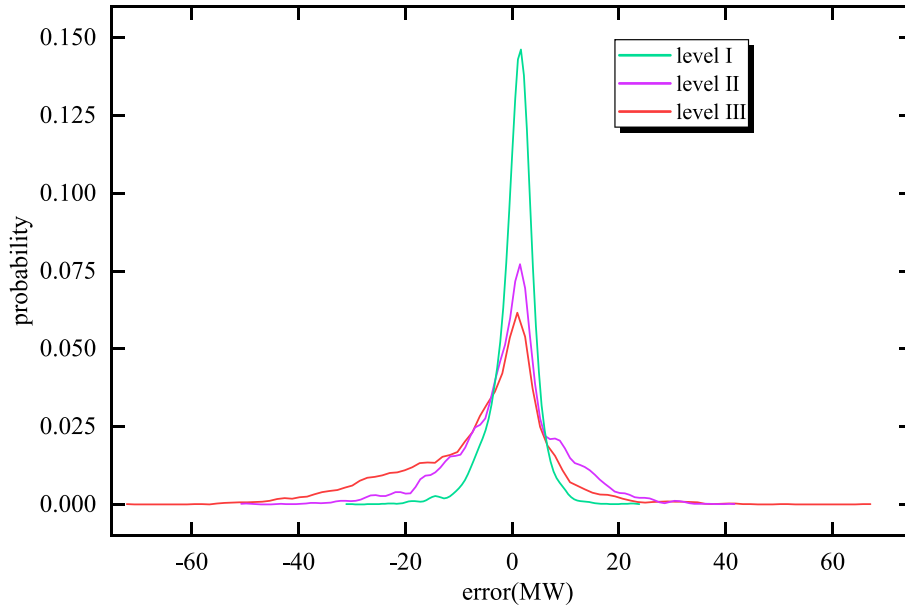


Fig. 2. The probability density distribution of wind power prediction error.

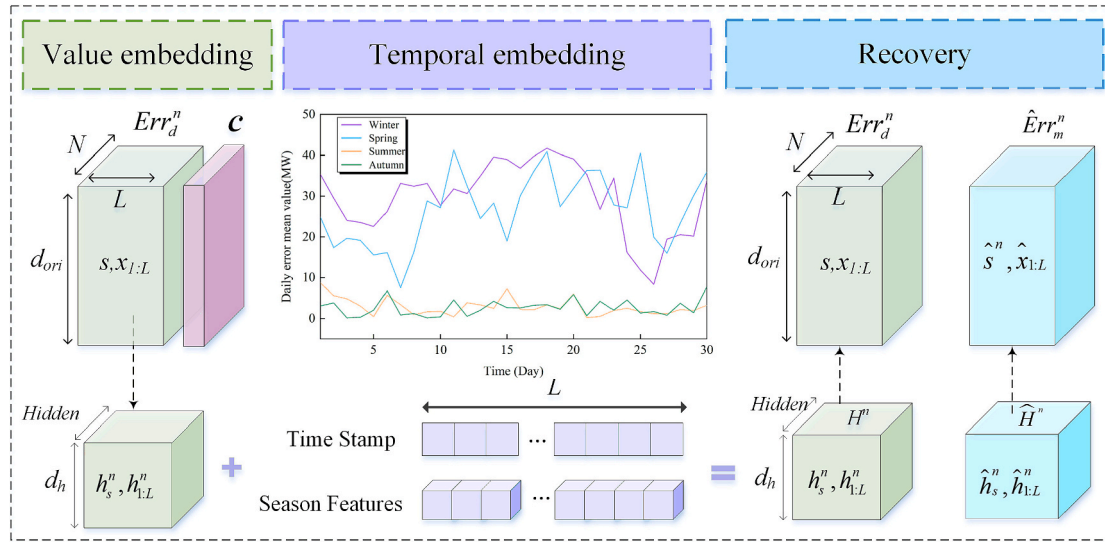


Fig. 3. The architecture of the Reconstruction block.

where R_s and R_t are the Recovery networks of static and dynamic features, respectively. $\tilde{h}_s^n, \tilde{h}_t^n$ denote the low-dimensional static and dynamic characteristics of real error $Err_n d$ or generation error scenarios $\hat{Err}_m^n, \tilde{s}^n$ and $\tilde{x}_t^n$ are the static and dynamic features after recovery to the original feature space.

2.4. Adversarial block

The Adversarial block accomplishes the wind power prediction error sequence generation task within the low-dimensional space H constructed by the Reconstruction block. The block consists of two deep neural networks, the generator and the discriminator, where the generator is used to learn the conditional probability mapping of the low-dimensional wind power prediction error set under label c . And the role of the discriminator is to extract the key features of the actual error eigenstate and the generation error eigenstate and differentiate them to output probability scores. Then feed the probability

scores to the discriminator and the generator to guide the network parameters for updated learning. The detailed composition of the generator and discriminator is described next, and its structure is shown in Fig. 4.

2.4.1. Generator based on temporal convolutional network

Considering that the wind power prediction error series has an apparent temporal correlation, i.e., there is a strong correlation between the data of adjacent periods, the generator should make full use of the information from the previous $t-1$ moments when generating the dynamic features of the prediction error at time t , and have the ability to capture the gradual change characteristics between the moments of the dynamic features. For this reason, this paper selects a temporal feature extraction model with autoregressive properties to build a generator network. The generators of existing time-series generative models are usually implemented based on RNN. However, the serial structure of RNN causes them to spend time and memory proportional to the length of the sequence, resulting in low computational efficiency of the

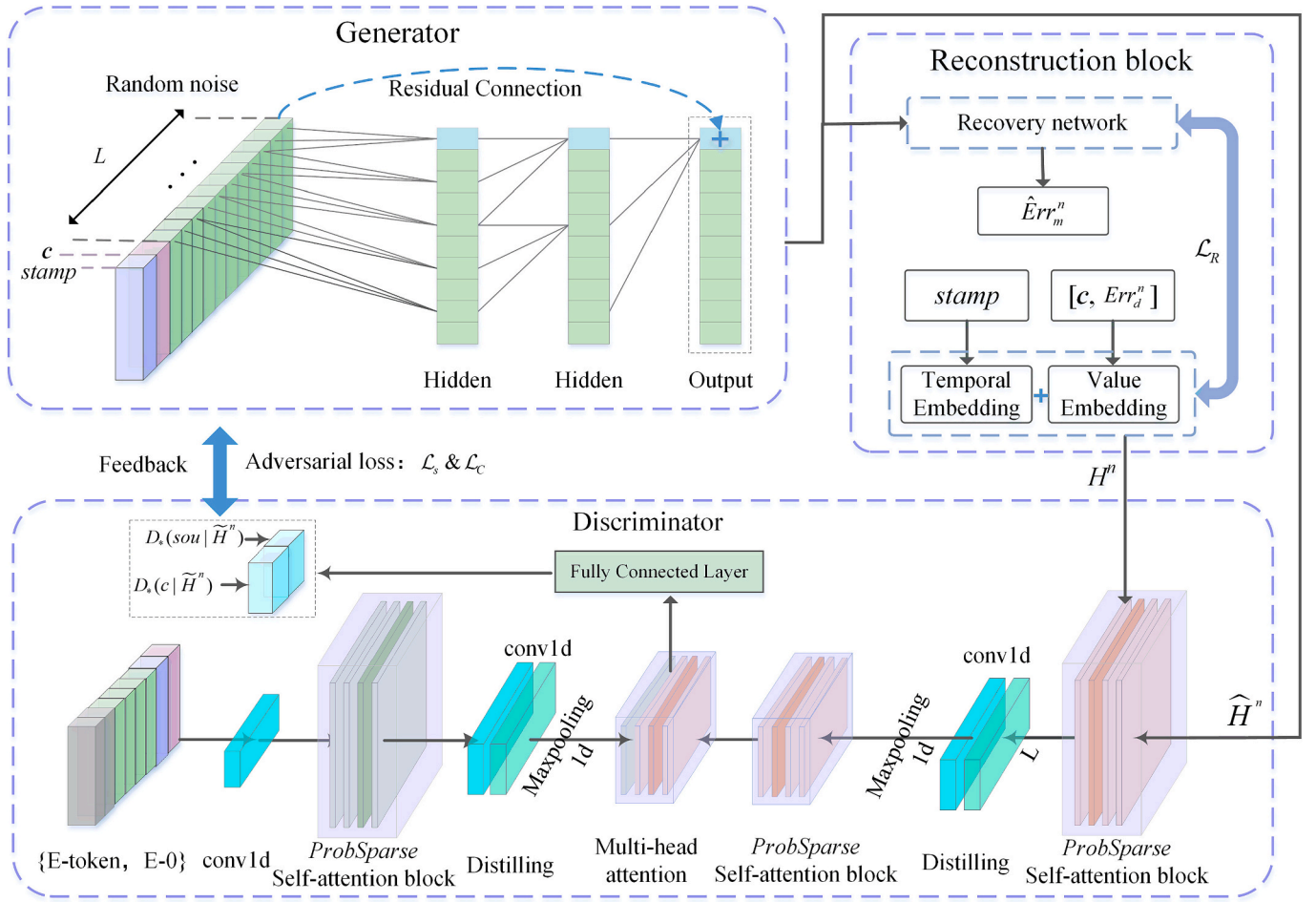


Fig. 4. Specific structure of the Adversarial block. (1) The generator part. The pink cuboid is the error level label c . The purple cuboid is the timestamp. The green cuboid input is a random noise vector with a length of L . (2) The discriminator part. The gray cuboid is the output discriminant result, which is filled with 0 elements when input. The green cuboid input by the discriminator is the real error sequence in the original space with a length of O . The lighter layers in the ProbSparse self-attention block are Q , K , and V matrices, and the darker layers are dot product matrices. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

generative models while easily ignoring local timing features. Unlike sequential models with serial structures such as RNN, the temporal convolutional network (TCN) uses causal convolution and dilated convolution structures to take into account both global and local temporal patterns, and reduces unnecessary neuron connections to increase training speed. Hence, we introduce the TCN into the generator network, aiming to use the special convolutional structure to do parallel computation on the timing information and mine the temporal features efficiently and adequately, so that the generator model can learn the conditional probability mapping of Err_d^n quickly and accurately. The random noise z of length L , the label c of the day-ahead prediction error, and the timestamp $stamp$ are combined as the input of the generator G^* , and the generated low-dimensional feature set $\hat{H}^n = \{\hat{h}_s^n, \hat{h}_{1:L}^n\}$ is output through the generator G^* . The generator can be expressed as

$$\begin{aligned} \hat{h}_s^n &= G_s(zs) \\ \hat{h}_t^n &= G_x(\hat{h}_s^n, \hat{h}_{t-1}^n, z_t, c, stamp) \end{aligned} \quad (4)$$

where G^* is the generator network, z^* is the static and dynamic features

of the random noise, and $\hat{h}_s^n$ is the static and dynamic feature vector generated by the generator.

2.4.2. Discriminator based on informer

The self-attention mechanism model can achieve attention to important information of sequence features by using weight assignment, and its application to the discriminator can enhance the classification discriminatory ability of the discriminator for real and generated error sequence features. The most widely used self-attention mechanism model in time series tasks is the Transformer, which can extract features of input data in parallel and model both long-term and short-term time series features. However, the secondary computation of its self-attention leads to high computational complexity and large memory usage. Besides, the stack of multiple encoders and decoders makes the total memory of the model large, which restricts its scalability when receiving long series input, so Transformer has some limitations in solving long time series problems. In the study of wind power scenario generation on the day-ahead time scale, the inputs for the model are long time sequences of wind power prediction error with a time resolution equal to

15 min and length equal to 96, and the performance of the Transformer in this application scenario will be greatly affected.

In response to the above deficiencies of Transformer, Informer proposes *ProbSparse* self-attention to replace the traditional self-attention mechanism by studying the sparsity of self-attention [33,34], thus reducing the computational complexity and memory usage of the model. In addition, Informer uses distillation operations for the stacked layers of encoders and decoders, reducing the spatial complexity and enabling the model to receive and handle long sequences efficiently. Therefore, in this paper, we construct a discriminator network based on Informer, and the discriminator consists of *ProbSparse* self-attention, Distilling, Multi-head attention, and Fully connected layer.

Step 1: Obtaining key features of long-time sequences by *ProbSparse* self-attention.

The real error series feature H^n and the generated error series feature $\hat{H}_n$ are multiplied by w_q , w_k , and w_v weight coefficients to form query(Q), key(K), and value(V) matrices, $Q \in R^{L_Q \times d_h}$, $K \in R^{L_K \times d_h}$, $V \in R^{L_V \times d_h}$, $L_Q = L_K = L_V = L$, d_h is the dimension of H^n and $\hat{H}_n$ data, and the dot-product operation between the matrices is performed to obtain the attention results between the data at each moment of the H^n and $\hat{H}_n$ series. Since the results of the self-attention scoring analysis present long-tail characteristics [35], i.e., only a small number of dot-product pairs have significant contributions, to distinguish the dot-product pairs that can produce significant contributions, it is necessary to do a sparsity measure calculation for the elements q_i of the Q matrix and define the i -th Q matrix sparsity measure as

$$M(q_i, K) = \ln \sum_{j=1}^L e^{\frac{q_i k_j^T}{\sqrt{d_h}}} - \frac{1}{L} \sum_{j=1}^L \frac{q_i k_j^T}{\sqrt{d_h}} \quad (5)$$

where $\frac{q_i k_j^T}{\sqrt{d_h}}$ is the computed dot-product of the i -th query and the j -th transposed key, and L is the length of the K matrix, whose size is consistent with the length of the day-ahead wind power error sequence.

Suppose the $M(q_i, K)$ of the i -th query is larger and its attention probabilities are more diverse. In that case, the computed dot-product pair has a higher probability of producing an important contribution to the self-attention result, i.e., the characteristics of the wind power prediction error data at the moment i are more correlated with the data at other moments of the sequence. The results of the sparsity measure are sorted to obtain the matrix of the U queries with the largest $M(q_i, K)$. The U queries are called active queries, and the remaining $L-U$ queries are called lazy queries. Each K gets self-attention only by U active queries.

$$\mathcal{A}(\bar{Q}, K, V) = \text{Softmax}\left(\frac{\bar{Q}K^T}{\sqrt{d}}\right)V \quad (6)$$

where $\bar{Q}$ consists of U active queries, $\mathcal{A}(\bar{Q}, K, V)$ is the self-attention calculation result of the corresponding row of active query, for lazy query corresponding row of self-attention score will be directly replaced by the mean value of V matrix, the two parts of self-attention results are stitched to obtain the self-attention feature F_p , the specific calculation process is shown in Fig. 5.

Step 2: Highlighting the dominant attention of F_p .

Considering the redundant information combination of the V matrix in the self-attention result F_p , we reduce the time dimension of the input sequence by the Distilling layer to further capture the most effective attention information F_h in F_p .

$$F_h = \text{Distilling}(F_p) = \text{MaxPool}(\text{ELU}(\text{Conv1d}(F_p))) \quad (7)$$

where Conv 1d is the convolutional layer, ELU is the activation function, and MaxPool is the maximum pooling layer.

Step 3: Feature interaction of original and latent space.

Using the *Err n d* sequences in the original feature space as excitation is beneficial for the discriminator to complete the distinction between true and false error features quickly, so in this paper, the *Err n d* fragments of length O are input into a new *ProbSparse* self-attention layer and Distilling layer to extract the key features F_o in the original feature space. Then the mutual attention results between F_o and F_h are calculated by Multi-head attention for information interaction. Finally, the discriminative results of data feature authenticity and label accuracy are output through the Fully connected layer. The following equation can represent the discriminator

$$[D_*(\text{sou}|\tilde{H}^n), D_*(c|\tilde{H}^n)] = f_{FC}(f_{AT}(f_{\text{prob}}(f_{\text{distill}}(f_{\text{prob}}(\tilde{H}^n)), \text{Err}_{1:O}^n))) \quad (8)$$

where $\tilde{H}$ is the general term for H^n and $\hat{H}_n$, f_{prob} , f_{distill} , f_{AT} , and f_{FC} correspond to the *ProbSparse* self-attention, Distilling, Multi-head attention, and Fully connected layer of the discriminator network. $D_*(\text{sou}|\tilde{H}^n)$ and $D_*(c|\tilde{H}^n)$ are the discriminatory results on the authenticity and label accuracy of the low-dimensional error feature set, *sou* denotes the data source.

2.5. Joint learning

This section introduces the loss function established by the Informer-TimeGANs model training, where the prediction error data of different strata are trained using the same generative model. Algorithm 1 gives the Informer-TimeGANs training pseudo-code, which summarizes the complete training process of Informer-TimeGANs, where E denotes the Value embedding and Temporal embedding functions with θ_e as the parameter, R denotes the Recovery function with θ_r as the parameter, G indicates the generator network with θ_g as the parameter, D indicates the discriminator network with θ_d as the parameter, and θ_e , θ_r , θ_g and θ_d are the weight parameters of these four deep neural networks, respectively.

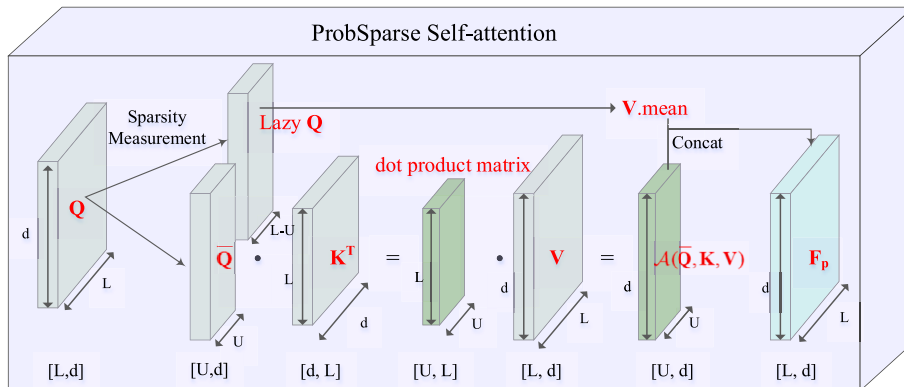


Fig. 5. The computational process of *ProbSparse* Self-attention.

Algorithm 1. Informer-time-series generative adversarial.

Require: Learning rate α , batch size b , Number of iterations for generator per discriminator iteration n_{gen}

Require: Initial weights θ_e for E, θ_r for R, θ_g for G and θ_d for D

```

1 for  $m_{iter}$  Reconstruction training iterations do
2   update embedding  $\theta_e$  and recovery  $\theta_r$ 
3   Sample batch samples  $\{Err_d^n\}_i^b$  from train set;
4   Update parameters for embedding and recovery network
5      $g_{\theta_e}, g_{\theta_r} \leftarrow \nabla \theta_g \frac{1}{b} \sum_{i=1}^b [\| (Err_d^n)_i - R(E(Err_d^n)_i) \|_2]$ 
6      $\theta_e \leftarrow \theta_e - \alpha \cdot Adam(\theta_e, g_{\theta_e}, \beta_1, \beta_2)$ 
7      $\theta_r \leftarrow \theta_r - \alpha \cdot Adam(\theta_r, g_{\theta_r}, \beta_1, \beta_2)$ 
8 end
9 for  $n_{iter}$  Adversarial training iterations do
10   for  $t = 0, \dots, n_{gen}$  do
11     update parameters for generator  $\theta_g$ 
12      $g_{\theta_g} \leftarrow \nabla \theta_g \frac{1}{m} \sum_{i=1}^m (\eta \mathcal{L}_{Sup} + \mathcal{L}_S - \mathcal{L}_C)$ 
13      $\theta_g \leftarrow \theta_g - \alpha \cdot Adam(\theta_g, g_{\theta_g}, \beta_1, \beta_2)$ 
14     update parameters for discriminator  $\theta_d$ 
15      $g_{\theta_d} \leftarrow \nabla \theta_d \frac{1}{m} \sum_{i=1}^m (\mathcal{L}_S + \mathcal{L}_C)$ 
16      $\theta_d \leftarrow \theta_d - \alpha \cdot Adam(\theta_d, g_{\theta_d}, \beta_1, \beta_2)$ 
17 end

```

The Reconstruction block constructs the Reconstruction loss by the two-norm between the static and dynamic features s^n and x^n in the original feature space of $Err_n d$ and the $\hat{s}^n$ and $\hat{x}_t^n$ recovered from the latent space, and continuously updates the parameters θ_e and θ_r during the training process to establish a reversible mapping between the latent space and the original feature space. The $\mathcal{L}_R$ expression is as follows

$$\mathcal{L}_R = E_{S, X_{1:L} \sim p} \left[\|s^n - \hat{s}^n\|_2 + \sum_t \|x_t^n - \hat{x}_t^n\|_2 \right] \quad (9)$$

In the Adversarial block, the training goal of the generator is to accurately capture the true $Err_n d$ joint probability distribution $P(h_n s, h_n 1:L)$ under the supervision of label c , such that the sequence probability distribution $\hat{P}(\hat{h}_s^n, \hat{h}_{1:L}^n)$ of the generated scenarios approximates $P(h_n s, h_n 1:L)$ as closely as possible, with the objective function of minimizing $\mathcal{L}_S - \mathcal{L}_C$. The training objective of the discriminator is to accurately describe the differences between the sample distributions and the correctness of the sample labels, and the objective function is to maximize $\mathcal{L}_S + \mathcal{L}_C$. Where the expressions of the adversarial discriminant loss function $\mathcal{L}_S$ and the auxiliary classification loss function $\mathcal{L}_C$ are as follows

$$\begin{aligned} \mathcal{L}_S = & \mathbb{E}_{s, x_{1:L} \sim p} \left[\log y_{(sou=real|h_s^n)} + \sum_t \log y_{(sou=real|h_t^n)} \right] \\ & + \mathbb{E}_{s, x_{1:L} \sim \hat{p}} \left[\log y_{(sou=fake|\hat{h}_s^n)} + \sum_t \log y_{(sou=fake|\hat{h}_t^n)} \right] \end{aligned} \quad (10)$$

Table 2
The Informer-TimeGANs network components in detail.

Reconstruction block:	
Embedding:	
Value embedding	(Dilated Causal Conv, Relu, Dropout)*2, residual connection
Temporal embedding	Embedding dim = d_h
Recovery:	
(Dilated Causal Conv, Relu, Dropout)*2, residual connection	
Adversarial block:	
Generator:	
[(Dilated Causal Conv, Relu, Dropout)*2, residual connection]*3	
Discriminator:	
ProbSpare self-attention block	Multi-head ProbSpare Attention n-heads = 4, d = d_h Add, LayerNorm, Dropout($p = 0.1$) Pos-wise FFN
Distilling	Conv 1d, ELU, Max pooling
Multi-head Attention	n-heads = 4, d = d_h
Full connection	d = d_{out}

$$\mathcal{L}_C = \mathbb{E}_{\mathbf{s}, \mathbf{x}_{1:T} \sim p} \left[\log y_{(C=c|\mathbf{s})} + \sum_i \log y_{(C=c|i)} \right] + \mathbb{E}_{\mathbf{s}, \mathbf{x}_{1:T} \sim \hat{p}} \left[\log y_{(C=c|\mathbf{s})} + \sum_i \log y_{(C=c|i)} \right] \quad (11)$$

where $y_{(sou=real|h \ n \ s/h \ n \ t)}$ denotes the probability that the discriminator will judge $h \ n \ s$ and $h \ n \ t$ as true, $y_{(sou=fake|\hat{h}_s^n/\hat{h}_t^n)}$ denotes the probability

that it will judge $\hat{h}_s^n$ and $\hat{h}_t^n$ as fake. $y_{(C=c|h \ n \ s/h \ n \ t)}$ represents the probability that the discriminator will judge the labels of $h \ n \ s$ and $h \ n \ t$ correctly and $y_{(C=c|\hat{h}_s^n/\hat{h}_t^n)}$ represents the probability that the discriminator will judge the labels of $\hat{h}_s^n$ and $\hat{h}_t^n$ correctly.

The above loss function contains unsupervised adversarial discriminant loss and auxiliary classification supervised loss. On top of this, an additional autoregressive supervised loss is introduced in this paper so that the generator receives the features of the previous $t-1$ moments $Err \ n \ d$ after the E network to generate the data at time t . The generator is further motivated to capture the stepwise dynamic conditional distribution in the wind power prediction error data. Finally, the objective function of the generator is updated to minimize $\eta \mathcal{L}_{sup} + \mathcal{L}_S - \mathcal{L}_C$, $\mathcal{L}_{sup}$ is defined as

$$\mathcal{L}_{sup} = E_{S, X_{1:T}} \left[\sum_i \|h_t^n - G_s(\hat{h}_s^n, \hat{h}_{1:t-1}^n, z_t, c, stamp)\|_2 \right] \quad (12)$$

3. Case analysis

3.1. Data description and experiment configuration

In order to validate the robustness of the proposed method, this paper evaluates the Informer-TimeGANs and the benchmark models using two datasets from different geographical locations. The two datasets are the day-ahead power prediction and prediction error data for the whole year of 2017 and from January 1, 2018, to November 31, 2018, in Northwest and Northeast China, respectively, with a temporal resolution of 15 min. 75% of the sample data is taken for training and the remaining 25% is used as a test set.

All the program for the Informer-TimeGANs model is implemented in the Python platform with Inter Xeon Platinum 8260 CPU at 3.90GHz and NVIDIA GeForce RTX 3090 GPU through PyTorch deep learning framework. Table 2 summarizes the structure and parameter details of Informer-TimeGANs.

3.2. Comprehensive evaluation metrics

In order to comprehensively evaluate the quality of the day-ahead wind power scenario set, the model-generated scenario set is analyzed from three perspectives: temporal correlation, volatility, and accuracy. Temporal correlation can reflect the actual operation of wind power in terms of time characteristics and avoid underestimation of operating costs [36]. In this paper, the time-series autocorrelation coefficient $R(\tau)$ is used to measure the correlation of time series in two different periods to assess the learning ability of the proposed model for time series features.

$$R(\tau) = \frac{E[(S_t - \mu)(S_{t+\tau} - \mu)]}{\sigma^2} \quad (13)$$

where S_t is the value of the generated error sample or the true error sample series at moment t , μ is the expected value of the sample series, σ is the standard deviation, and τ is the time lag.

The volatility of wind power is an important feature of the wind power time series change pattern. The volatility of wind power is defined as P_{Vol} , and the volatility interval of the day-ahead wind power scenario set is P_t , which is designed to characterize the accuracy of the

day-ahead wind power scenario set in describing the fluctuation level of real wind power.

$$P_{Vol} = \frac{\sum_{t=1}^L |P_{t+1} - P_t|}{P_N} \quad (14)$$

$$P_t = [P_{Vol}^{\min}, P_{Vol}^{\max}] \quad (15)$$

where P_{t+1} and P_t are the wind power values of the wind farm at the moment $t+1$ and t , respectively, L is the number of day-ahead wind power moments, and P_N is the installed capacity of the wind farm. $P_{\min}$ and $P_{\max}$ are the lower and upper limits of the volatility of the day-ahead wind power scenario set, respectively.

For the accuracy assessment of the scenario set, this paper adopts coverage rate (CR) and power interval average width (PIW) as evaluation metrics, where CR is used to evaluate the coverage of the confidence interval of the scenario set to the measured values, and PIW is used to assess the ability of the generated day-ahead scenario set to gather uncertain information. Specifically, when the CR of the scenario set is larger and the PIW is smaller, its description of power uncertainty is more accurate, and if a similar CR exists, the smaller the PIW is, the better the accuracy of the model.

$$CR_{(1-\alpha)} = \left[\frac{N_{(1-\alpha)}}{L} \right] \times 100\% \quad (16)$$

where $CR_{(1-\alpha)}$ is the CR value at the confidence level $1-\alpha$, $N_{(1-\alpha)}$ is the number of measured power values falling into the confidence interval of the wind power scenario set at the confidence level $1-\alpha$.

$$PIW_{(1-\alpha)} = \frac{1}{T} \sum_{t=1}^T (P_{t,1-\alpha}^{\max} - P_{t,1-\alpha}^{\min}) \quad (17)$$

where $PIW_{(1-\alpha)}$ is the PIW value at the confidence level $1-\alpha$, $P_{t,1-\alpha}^{\max}$ and $P_{t,1-\alpha}^{\min}$ are the maximum and minimum values of the wind power scenario set at time t under the confidence level $1-\alpha$.

3.3. Temporal correlation analysis

In addition to Informer-TimeGANs proposed in this paper, four benchmark models named GAN, RNN-GAN, Seq-GAN, and TimeGAN are used for comparison. The above benchmark models are all set with the proposed Error stratification block in this paper.

Multiple randomly selected samples from different levels of prediction error data in the test set are analyzed, and input the random noise z and error level c into the trained generator. The autocorrelation coefficient $R(\tau)$ variation curves of the error scenarios and day-ahead wind power scenarios generated by different models for one set of samples compared with the real data are given in Figs. 6 and 7, respectively. The coefficients are calculated at intervals from 0 to 7 h.

From the overall characteristics of the three data sets demonstrated in Fig. 6 and Fig. 7, the $R(\tau)$ of the generated error series and the wind power scenario set gradually decrease with the increase of the time interval, which is consistent with the time-series autocorrelation characteristic of the strong error and power correlation of the adjacent moments of renewables. This correlation will gradually weaken over time, the $R(\tau)$ will appear to be negative at larger time intervals, and the relationship between the data is negatively correlated, due to the significant change in the characteristics of the error and power data at larger time intervals.

According to the results of different error levels, in Level I small fluctuation strata, there are not too many sudden large fluctuations in the data, so its autocorrelation is more obvious. The gap between the effectiveness of the benchmark models in fitting temporal dependencies and Informer-TimeGANs is not very large. However, it can still be seen that GAN is difficult to learn the global correlation of sequences, and has

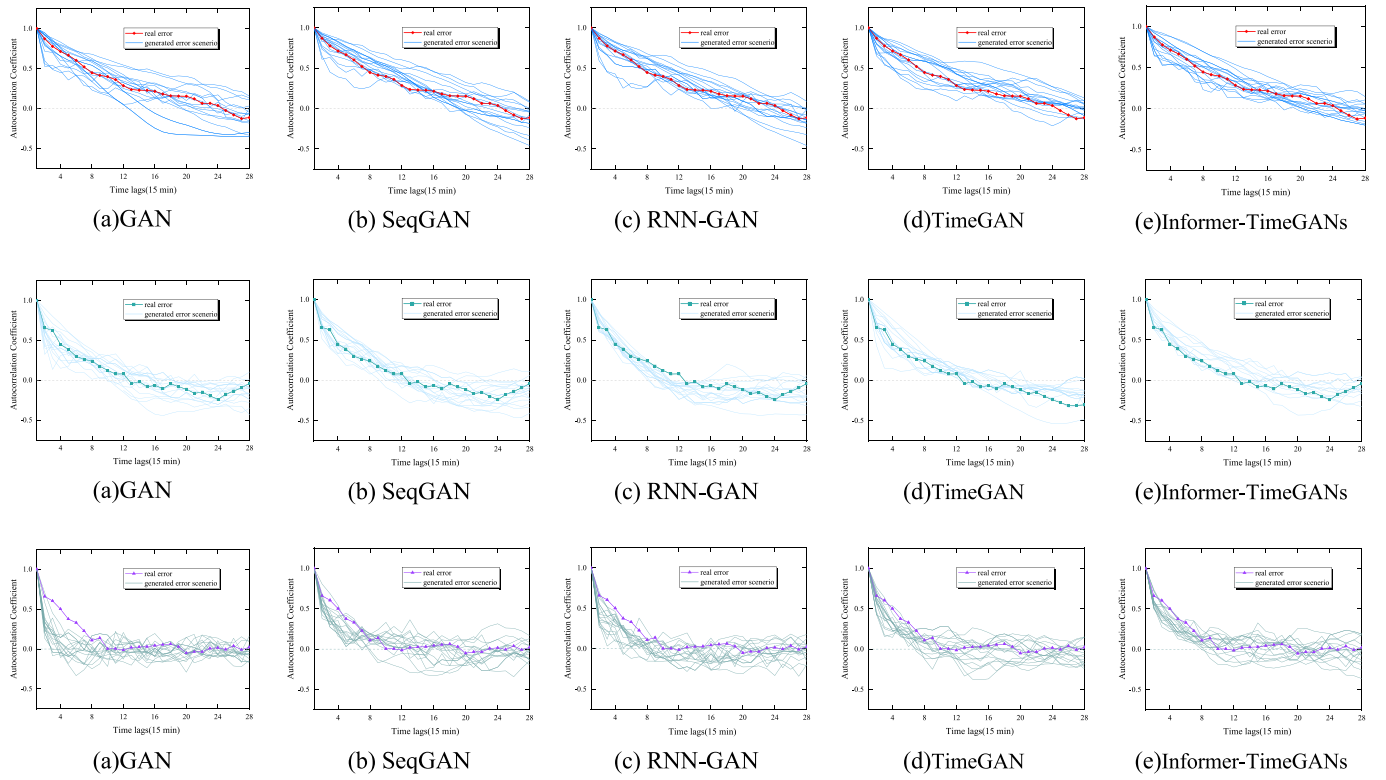


Fig. 6. Comparison of actual error and generated error scenarios using different methods.

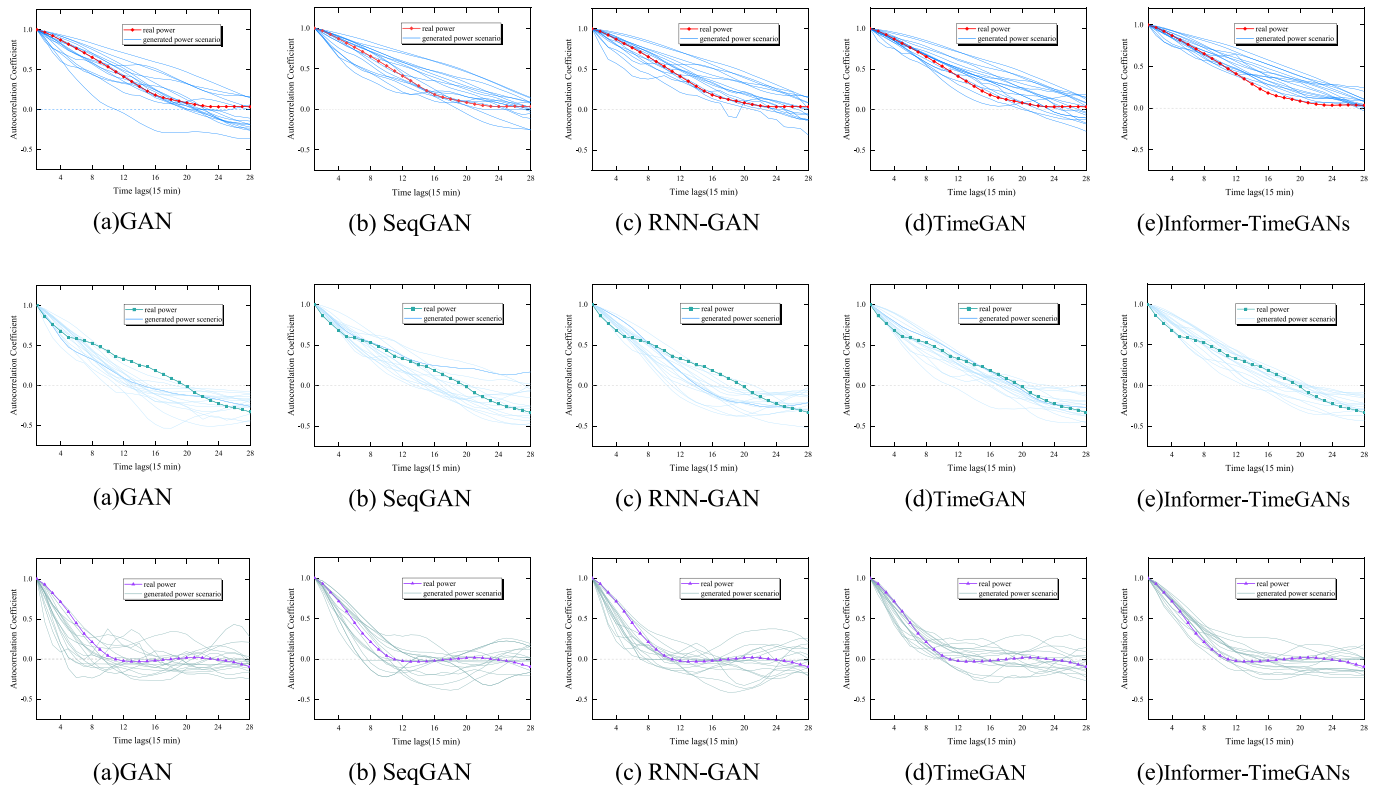


Fig. 7. Comparison of actual and generated wind power scenarios using different methods.

not accurately mined the correlations with sequence time intervals of 3 h or more. Although SeqGAN and RNN-GAN have embedded a temporal feature extraction layer compared with GAN, which enhances the

learning ability of time series autocorrelation. But it can also be found that the local error of some scenarios is large and the stability of learning for long time dependencies is poor.

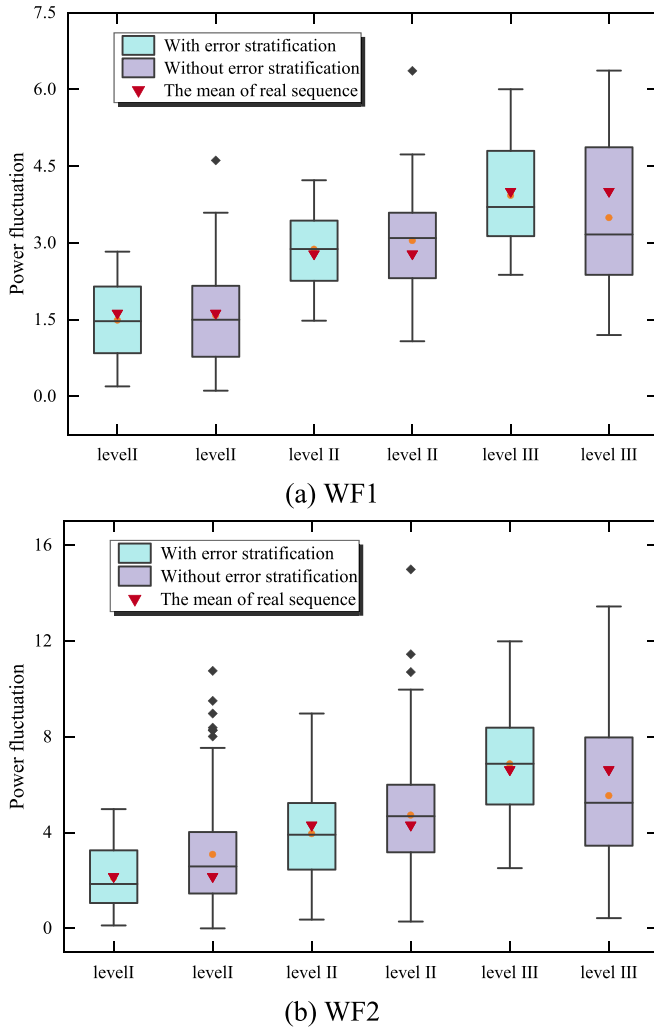


Fig. 8. The fluctuation box line diagram of the day-ahead scenarios of two wind farms.

Table 3
Comparison of P_{Vol} and P_I before and after error stratification.

Wind farm	Error level	Real	Informer-TimeGANs (With error stratification)			Informer-TimeGANs (Without error stratification)	
		P_{Vol}	P_{Vol}	P_I		P_{Vol}	P_I
WF1	Level I	1.63	1.49	[0.20,2.83]		1.59	[0.11,4.61]
	Level II	2.79	2.88	[1.48,4.23]		3.04	[1.08,6.36]
	Level III	4.01	3.93	[2.31,6.01]		3.49	[1.20,6.37]
WF2	Level I	2.16	2.10	[0.12,4.98]		3.08	[0.04,10.76]
	Level II	4.32	3.95	[1.01,7.05]		4.72	[0.28,14.99]
	Level III	6.63	6.88	[3.08,11.99]		5.54	[0.53,13.45]

Table 4
The P_{Vol} and P_I of the generated scenario set using different models with the error division block.

Wind farm	Error level	Real	GAN		Seq-GAN		RNN-GAN		TimeGAN		Informer-TimeGANs	
		P_{Vol}	P_{Vol}	P_I	P_{Vol}	P_I	P_{Vol}	P_I	P_{Vol}	P_I	P_{Vol}	P_I
WF1	Level I	1.63	1.70	[0.19,6.37]	2.23	[0.56,4.91]	2.02	[0.17,4.93]	1.48	[0.08,3.67]	1.49	[0.20,2.83]
	Level II	2.79	3.54	[0.82,6.80]	2.69	[0.91,7.17]	2.55	[0.41,4.89]	2.92	[1.34,4.99]	2.88	[1.48,4.23]
	Level III	4.01	4.30	[1.45,8.30]	3.48	[1.61,5.27]	3.31	[0.77,6.91]	3.55	[2.19,6.35]	3.93	[2.31,6.01]
WF2	Level I	2.16	3.05	[0.36,10.76]	1.82	[0.18,10.76]	1.90	[0.05,7.30]	2.12	[0.19,7.68]	2.10	[0.12,4.98]
	Level II	4.32	5.83	[3.61,10.97]	5.22	[0.73,9.50]	5.04	[2.24,10.37]	5.56	[2.59,9.51]	3.95	[1.01,7.05]
	Level III	6.63	7.02	[2.93,14.99]	6.34	[1.48,10.33]	6.06	[2.84,10.70]	6.94	[4.25,13.45]	6.88	[3.08,11.99]

As the degree of data fluctuation rises, the autocorrelation gap of the scenarios generated by benchmark models and Informer-TimeGANs becomes gradually obvious in the data with higher fluctuations in Level II and Level III. In particular, the overall fluctuation of data in Level III is drastic, and the error data only have a significant correlation with the data of a few moments close to each other. The real error and measured power data show no obvious autocorrelation after a time interval of 2 h and 2.5 h, respectively. It is difficult for the benchmark models to extract the stochastic fluctuation characteristics of the series from the complex and rapidly changing wind power prediction error data, resulting in a low overall accuracy of the autocorrelation of the generated scenarios.

Both in terms of the overall pattern of variation in the timing autocorrelation properties and terms of the physical significance contained in the different error levels. The day-ahead scenarios generated by Informer-TimeGANs can maintain a high degree of consistency with the timing autocorrelation characteristics of actual power data, and accurately reproducing the temporal characteristics of wind power data. This is attributed to the balance between global and local features of the time series by TCN and the extraction of key fluctuation features of the series by Informer.

3.4. Volatility analysis

Fig. 8 shows the box line plot of the volatility P_{Vol} in the generation scenario sets from two wind farms, where the red inverted triangle symbols are the mean values of the measured wind power data volatility, the orange circular symbols are the mean values of the scenario set volatility, and the black diamond symbols are the outliers. It can be seen that the scenario set generated by the model without the Error stratification block is more conservative, its P_{Vol} has some outliers and the boundaries between P_{Vol} of each error layer scenario are not clear, which describes the fluctuation characteristics of the day-ahead wind power as relatively vague. After setting the Error stratification block, not only the difference between the mean volatility value of the generated scenarios and the actual wind power is smaller, and the volatility interval P_I of the scenario set is narrower, but also the distribution of P_{Vol} values among the scenario sets of each error level is significantly different, which can describe the fluctuation characteristics of wind power more accurately. Table 3 presents the statistical results of P_{Vol} and P_I for the day-ahead wind power scenarios generated by Informer-TimeGANs before and after setting the Error stratification block.

In order to better validate the performance of the proposed method in simulating the volatility of real data, the P_{Vol} and P_I results of the benchmark models after setting the error division and the proposed model in this paper for generating scenarios are compared, as shown in Table 4. It can be found that the Informer-TimeGANs method generates day-ahead scenarios that are more similar to the fluctuation characteristics of the measured power data in each error level. Although the average volatility of the scenarios generated by the GAN and TimeGAN methods are closest to the measured power data in the small fluctuation Level I of the two datasets, respectively. Yet the upper bounds of their volatility intervals are not consistent with the characteristics of the small volatility layer, indicating that the accuracy of the model in simulating

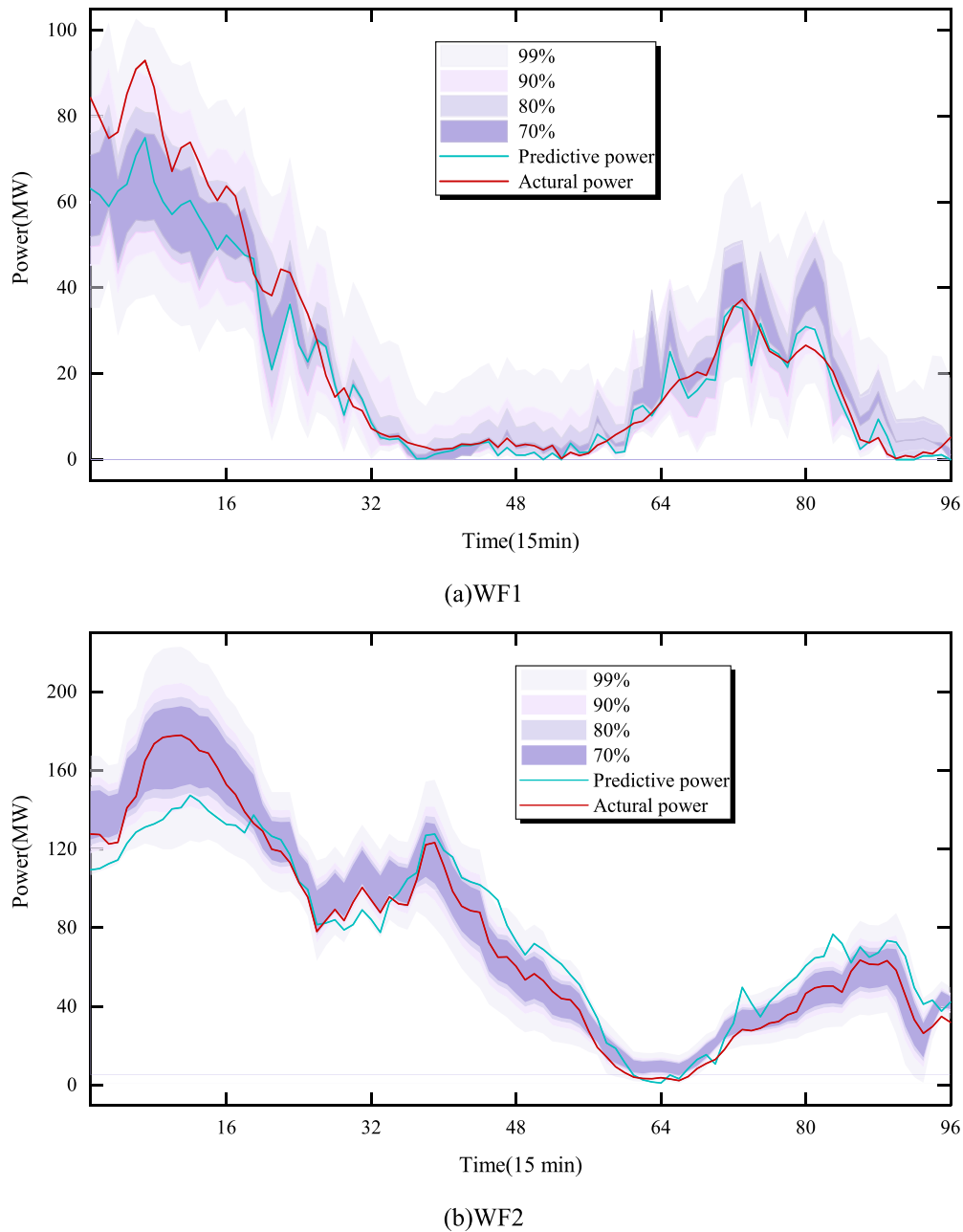


Fig. 9. The day-ahead scenario set power confidence interval of Dataset1.

the volatility of real data is not stable enough.

3.5. Accuracy analysis of different methods

Figs. 9 and 10 show power confidence intervals of the day-ahead scenario sets constructed by Informer-TimeGANs for wind farms under two datasets. As can be seen in the figure, The confidence interval of prediction error is larger in the high-power interval and the period with strong fluctuations. In comparison, the confidence interval of prediction error is smaller in the low-power interval or the period with a low level of power fluctuations. This demonstrates that the day-ahead scenario set generated by Informer-TimeGANs can effectively describe the error characteristics of wind power at different moments.

In this paper, 50-day samples are randomly selected from within the test set of two datasets to avoid the lack of rigor in validating a single sample. Table 5 gives the statistical results of the CR and PIW for the day-ahead scenario set generated by Informer-TimeGANs, GAN,

SeqGAN, RNN-GAN, and TimeGAN. From a single metric, there are cases in the benchmark models where CR is higher than Informer-TimeGANs or PIW is lower than Informer-TimeGANs. However, high CR in these cases is always combined with large PIW. This suggests that the distribution of scenarios is relatively decentralized, with a high degree of coverage of the measured data yet not accurately describing the upper and lower bounds of uncertainty. Of course, there are also combinations of low CR and small PIW, and the reason for the low level of coverage of real power and narrow PIW is that the model undergoes a certain degree of pattern collapse, which results in a relatively homogenous set of scenarios for its generation.

In summary, the scenario set generated by the model proposed in this paper has the most accurate portrayal of the uncertainty of day-ahead wind power, and its distribution is more concentrated while ensuring the CR of the measured power. This shows that Informer-TimeGANs have excellent performance in the day-ahead power generation task, and the superior performance of the model is mainly attributed to the

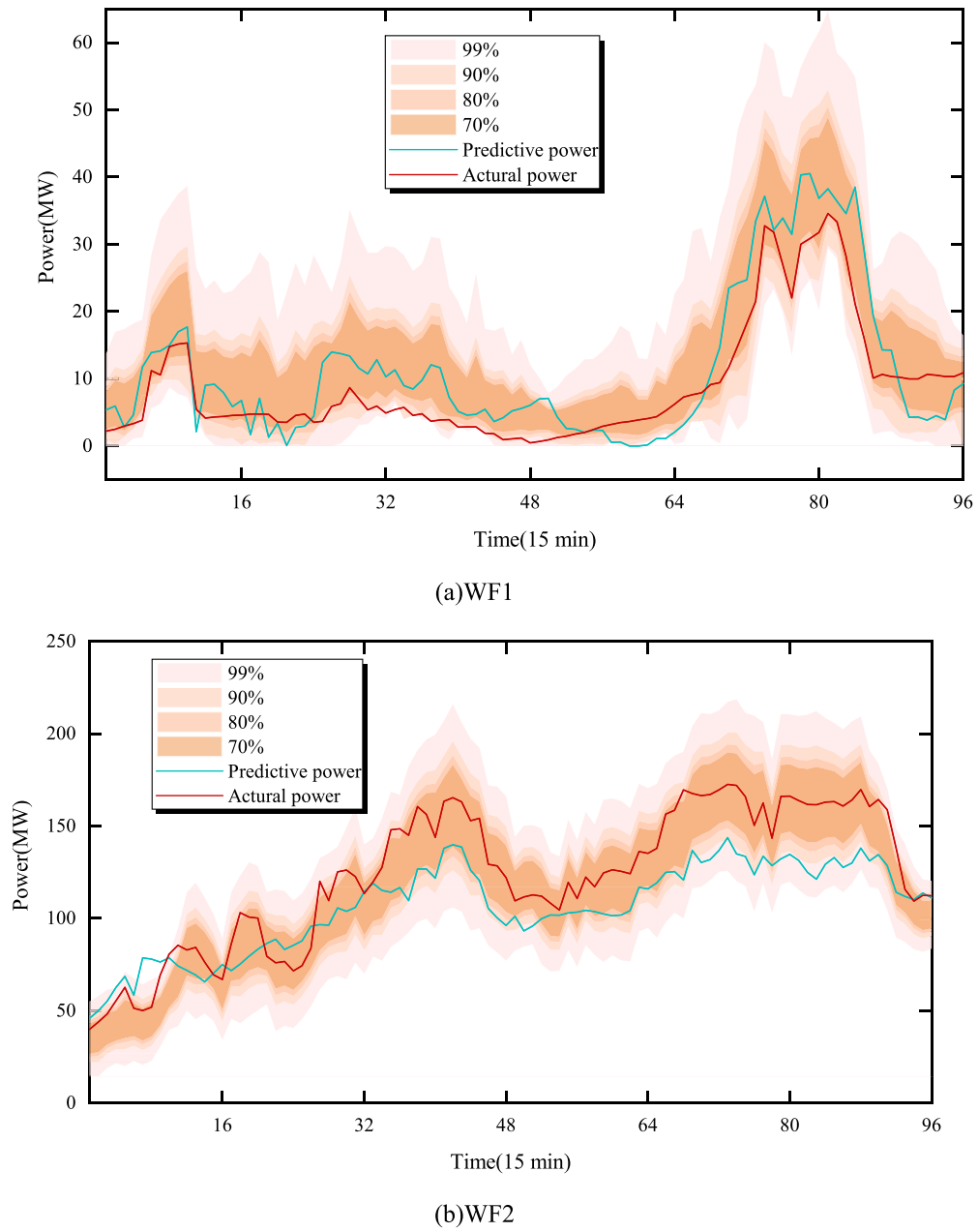


Fig. 10. The day-ahead scenario set power confidence interval of Dataset 2.

Table 5

Accuracy comparison of generated scenarios between Informer-TimeGANs and benchmark models.

Wind farm	Confidence degree	GAN		SeqGAN		RNN-GAN		TimeGAN		Informer-TimeGANs		
		CR/%	PIW/MW	CR/%	PIW/MW	CR/%	PIW/MW	CR/%	PIW/MW	CR/%	PIW/MW	
Dataset 1	WF1	99%	80.90	45.39	91.04	43.24	89.97	44.63	96.86	42.39	<u>98.34</u>	35.07
		90%	70.22	35.96	84.97	32.83	78.44	34.23	86.24	31.36	<u>92.71</u>	20.85
		80%	60.09	30.59	<u>73.96</u>	25.57	68.55	28.67	61.37	24.75	72.92	<u>14.56</u>
	WF2	99%	88.54	61.25	90.63	70.09	87.50	<u>49.24</u>	94.79	61.49	<u>96.88</u>	55.50
		90%	66.67	43.81	71.88	44.56	79.17	<u>34.31</u>	71.88	39.09	<u>85.42</u>	35.28
		80%	55.21	34.12	62.50	30.43	72.92	28.97	59.37	30.51	<u>80.21</u>	<u>27.53</u>
Dataset 2	WF1	99%	87.88	40.39	92.36	38.29	<u>98.04</u>	35.27	93.22	38.95	96.88	<u>24.39</u>
		90%	72.22	26.95	82.29	20.13	78.44	25.61	80.90	17.34	<u>86.46</u>	<u>15.39</u>
		80%	62.50	23.45	68.23	<u>11.97</u>	66.85	15.22	73.96	13.01	<u>79.17</u>	12.06
	WF2	99%	97.17	90.25	91.38	82.07	92.36	87.96	95.13	83.31	<u>97.92</u>	75.49
		90%	82.29	66.78	88.71	61.70	90.14	53.44	85.07	57.22	<u>90.63</u>	47.98
		80%	72.43	<u>36.75</u>	81.79	46.18	78.44	48.16	75.69	43.63	<u>84.38</u>	37.45

Table 6
Scenario accuracy index comparison of different models.

Wind farm	Confidence degree	LSTM-TimeGAN		Transformer-TimeGANs		Informer- TimeGANs	
		CR/%	PIW/MW	CR/%	PIW/MW	CR/%	PIW/MW
Dataset 1	WF1	99%	98.59	39.98	95.79	34.89	98.34
		90%	90.97	25.36	90.14	20.71	20.85
		80%	73.96	18.43	70.98	14.72	14.56
	WF2	99%	95.79	60.18	97.38	58.34	96.88
		90%	85.99	38.77	84.09	33.78	35.28
		80%	80.07	28.94	81.27	28.02	80.21
Dataset 2	WF1	99%	97.42	29.76	95.13	22.72	96.88
		90%	86.54	19.36	85.76	17.16	86.46
		80%	75.26	14.35	73.96	11.24	12.06
	WF2	99%	97.17	76.59	97.38	73.47	75.49
		90%	88.54	48.79	88.71	50.83	47.98
		80%	81.79	37.99	81.60	35.26	37.45

fact that Informer has fully explored the features with different contributions in the day-ahead error series and given targeted attention to them. GAN has the worst overall performance due to the lowest adaptation of its internal structure to the time series. The accuracy of scenarios generated by SeqGAN and RNN-GAN is improved to a certain extent relative to the GAN, which is due to the fact that they combine recurrent neural networks with GAN networks, and the performance of the extraction of temporal features rises. However, its generation process relies only on adversarial training, with insufficient learning of the seasonal characteristics and temporal gradual dynamic change characteristics of wind power, resulting in a general accuracy of the generated scenarios.

Furthermore, we keep the other blocks of the model unchanged and only replace the discriminator block with LSTM and Transformer to obtain LSTM-TimeGAN and Transformer-TimeGANs models as a set of control models for validating the effectiveness of Informer. Table 6 gives the accuracy metrics of the scenarios generated by the control models and the method proposed in this paper for both datasets.

LSTM-TimeGAN generates a portion of the scenario with a large difference from the real error sequence, which somewhat increases the CR to the measured power sequence but also makes the PIW wider. In the case where the difference in the CR of the three methods for generating scenarios to measure power is small, we believe that the effectiveness of the LSTM-TimeGAN model with a wider PIW is relatively poor. Further synthesizing the CR and PIW results of the two datasets, Informer-TimeGANs outperforms the Transformer-TimeGANs method. This is attributed to the ability of *ProbSparse* self-attention to focus more on the valid information of the sequence, which improves the ability of the model to discriminate between real and generated data, and in the adversarial game makes the generator enhances the learning of the real data distribution.

4. Conclusion

In this paper, a novel time-series generation adversarial network model is proposed to build the adversarial block of the generation model based on the TCN network and Informer structure to learn the dynamic characteristics of the wind power prediction error series, and construct an accurate and effective day-ahead scenario set of wind power. The error stratification block and the temporal embedding function are designed to improve the controllability of the scenario generation, and an accurate and effective day-ahead scenario set for the wind power is constructed. The quality of the generated scenarios is evaluated from three aspects: time-series correlation characteristics, fluctuation characteristics, and accuracy. The results show that the method can generate scenarios with consistent time-series variation characteristics and fluctuation with the real data, ensuring a high coverage rate while accurately portraying the day-ahead wind power uncertainty interval.

Extreme events have a large impact on the safe and stable operation

of power systems. In the future, we will consider conducting research on scenario generation under extreme events, while adding a spatial feature extraction block to the generation model to further improve the model structure to consider more spatially relevant information for multi-wind farm scenario generation.

CRediT authorship contribution statement

Lin Ye: Supervision, Validation, Resources, Writing – review & editing. **Yishu Peng:** Conceptualization, Methodology, Software, Data curation, Visualization, Investigation, Writing – original draft. **Yilin Li:** Writing – review & editing. **Zhuo Li:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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